

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Time: 2 hours

Paper reference **8MA0/01**

Mathematics
Advanced Subsidiary
PAPER 1: Pure Mathematics

You must have:
 Mathematical Formulae and Statistical Tables (Green), calculator

Total Marks

Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear.
Answers without working may not gain full credit.
- Inexact answers should be given to three significant figures unless otherwise stated.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- There are 15 questions in this question paper. The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

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1. The curve C has equation

$$y = \frac{3}{2}x^3 - 5x - \frac{10}{x}$$

- (a) Find $\frac{dy}{dx}$ giving your answer in simplest form.

(3)

Given that the point $P(-2, 3)$ lies on C ,

- (b) find the equation of the normal to C at P , giving your answer in the form $ax + by + c = 0$ where a , b and c are integers.

(4)



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Question 1 continued

Handwriting practice area with horizontal lines.

(Total for Question 1 is 7 marks)



2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Use algebra to solve the inequality

$$3x^2 > 7x + 10$$

Write your answer in set notation.

(3)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 3 marks)



3.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

$$f(x) = 6x^3 - (2a + 5)x^2 + 21x + a$$

where a is a constant.

Given that $(x - 2)$ is a factor of $f(x)$

(a) show that $a = 10$ (2)

(b) Hence solve the equation

$$f(x) = 0$$
 (4)

(c) Write down the largest solution of $f(-x) = 0$ (1)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 7 marks)



4. Given that $\vec{PQ} = -3\mathbf{i} + 12\mathbf{j}$

(a) find $|\vec{PQ}|$, giving your answer in simplest form.

(2)

Given also that relative to a fixed origin O

- the point P has position vector $9a\mathbf{i} - 4b\mathbf{j}$
- the point Q has position vector $6b\mathbf{i} + 15a\mathbf{j}$

where a and b are constants,

(b) find the value of a and the value of b .

(3)



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Question 4 continued

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(Total for Question 4 is 5 marks)



5.

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(i) Solve the equation

$$3(x - 2\sqrt{5}) = x\sqrt{5}$$

giving your answer in simplest form.

(3)

(ii) Find the exact value of x for which

$$e^{4x-1} = 5e^{\frac{1}{2}x}$$

(3)



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Question 5 continued

Lined area for writing the answer to Question 5.

(Total for Question 5 is 6 marks)



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Question 6 continued

Lined area for writing the answer to Question 6.



Question 6 continued

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Question 6 continued

Lined area for writing the answer to Question 6.

(Total for Question 6 is 7 marks)



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7. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$\left(2 - \frac{3}{4}x\right)^7$$

giving each term in simplest form.

(4)

Given that

$$f(x) = (x + 5)\left(2 - \frac{3}{4}x\right)^7$$

- (b) find the coefficient of x^2 in the expansion of $f(x)$

(2)



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Question 7 continued

Handwriting practice area with horizontal lines.

(Total for Question 7 is 6 marks)



8.

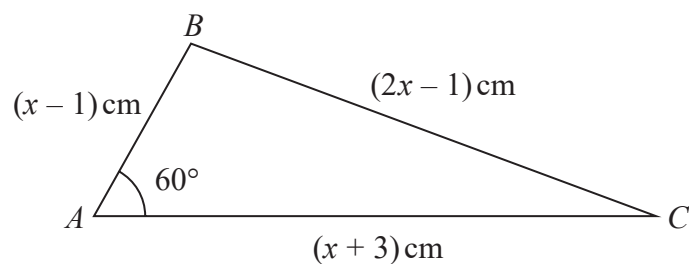


Figure 1

Figure 1 shows a sketch of triangle ABC .

Given that

- $AB = (x - 1) \text{ cm}$
- $BC = (2x - 1) \text{ cm}$
- $AC = (x + 3) \text{ cm}$
- $\text{angle } BAC = 60^\circ$

(a) show that $x^2 - 2x - 4 = 0$

(3)

(b) Hence find the area of triangle ABC giving your answer to 3 significant figures.

(3)



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Question 8 continued

Lined area for writing the answer to Question 8.

(Total for Question 8 is 6 marks)



9.

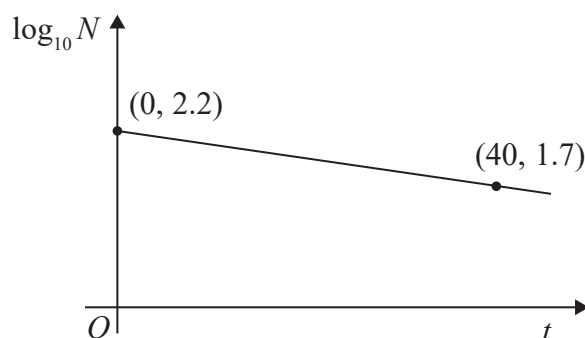


Figure 2

The number of fish in a lake is being monitored.

The line shown in Figure 2 models the linear relationship between $\log_{10} N$ and t , where

- N is the number of fish in thousands
- t is the number of years after monitoring began

The line passes through the points $(0, 2.2)$ and $(40, 1.7)$

Using this information,

- (a) find an equation for this line, (2)
- (b) find a complete equation for the model in the form

$$N = ab^t$$

where a and b are constants.

Give the value of a and the value of b , each to 3 significant figures. (3)

With reference to the model interpret,

- (c) (i) the value of a
- (ii) the value of b (2)
- (d) Find, according to the model, the number of fish when $t = 10$, giving your answer to the nearest 1000 (1)

The model predicts that T years after monitoring began, the number of fish will fall below 20 000 for the first time.

- (e) Find the value of T , giving your answer to the nearest integer. (2)
- (Solutions relying entirely on calculator technology are not acceptable.)*
- (f) Give a reason why the model may not be realistic. (1)



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Lined area for writing.



Question 9 continued

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Question 9 continued

Handwriting practice area with horizontal lines.

(Total for Question 9 is 11 marks)



10.

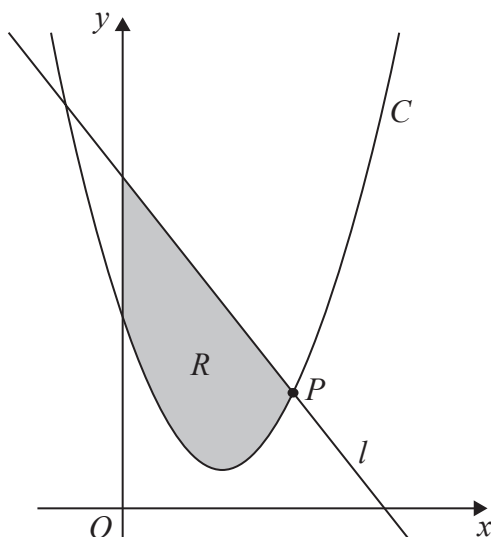


Figure 3

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

Figure 3 shows a sketch of the curve C with equation

$$y = 2x^2 - 7x + 8$$

and a sketch of the line l with equation

$$y = -3x + 14$$

The line l intersects the curve C at the point P as shown in Figure 3.

- (a) Find the x coordinate of P (3)

The region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

- (b) Use algebraic integration to find the area of R (4)



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Question 10 continued

Lined area for writing the answer to Question 10.



Question 10 continued

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Question 10 continued

Handwriting practice area with horizontal lines.

(Total for Question 10 is 7 marks)



11.

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

The furthest distance a particular electric car can travel at a constant speed, starting from a full charge, is modelled by the equation

$$y = \frac{1}{250}x^3 - \frac{3}{5}x^2 + 24x \quad 0 \leq x \leq 70$$

where y is the furthest distance in miles and x is the constant speed in miles per hour.

Use the model to answer parts (a), (b), (c) and (d).

- (a) Find the furthest distance the car can travel at a constant speed of 50 miles per hour, starting from a full charge. (1)
- (b) Use calculus to find the value of x for which y is a maximum, giving your answer to 3 significant figures. (4)
- (c) Using $\frac{d^2y}{dx^2}$ prove that y is maximised for the value of x found in part (b). (2)
- (d) Hence find the maximum distance the car can travel at a constant speed, starting from a full charge, giving your answer to the nearest mile. (2)



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Question 11 continued

Lined area for writing the answer to Question 11.



Question 11 continued

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Question 11 continued

Lined area for writing answers.

(Total for Question 11 is 9 marks)



12.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

Use the laws of logarithms to solve

$$4(\log_x 3 + \log_6 \sqrt{3}) = 3 \log_x 3 - \log_6 4$$

giving your answer as a simplified surd.

(5)



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Question 12 continued

Handwriting practice area with horizontal lines.

(Total for Question 12 is 5 marks)



13.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Prove that

$$\frac{\tan \theta (16 + 9 \sin^2 \theta)}{3 \sin \theta + 5 \tan \theta} \equiv 5 - 3 \cos \theta \quad \theta \neq (90n)^\circ \quad n \in \mathbb{Z} \quad (4)$$

(b) Hence solve for $0 < x < 180^\circ$

$$\frac{\tan 2x (16 + 9 \sin^2 2x)}{3 \sin 2x + 5 \tan 2x} = \frac{4}{\cos 2x} - 3$$

giving your answers to one decimal place.

(4)



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Question 13 continued

Lined area for writing the answer to Question 13.



Question 13 continued

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Question 13 continued

Handwriting practice area with horizontal lines.

(Total for Question 13 is 8 marks)



14. (a) Find

$$\int \frac{5x(5x-6)-9}{2\sqrt{x}} dx$$

giving each term in simplest form.

(4)

Given that

$$\int_1^k \frac{5x(5x-6)-9}{2\sqrt{x}} dx < 6\sqrt{k} + 14$$

where k is a constant, $k > 2$

(b) (i) show that

$$k^2 - 2k - 3 < 0$$

(ii) Hence find the range of values of k

(5)



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Question 14 continued

Lined area for writing the answer to Question 14.



Question 14 continued

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Question 14 continued

Lined area for writing the answer to Question 14.

(Total for Question 14 is 9 marks)



15. Prove, using algebra, that for all $x \in \mathbb{R}$

$$(x + 7)(x + 1) > 2x - 3$$

(4)



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Question 15 continued

Lined area for writing the answer to Question 15.



Question 15 continued

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(Total for Question 15 is 4 marks)

TOTAL FOR PAPER IS 100 MARKS

